MA4110/MA5020: Computational Methods for Fluid Flow

Lecture 8: Modified Equation Analysis: Numerical Dissipation and Dispersion

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1 Introduction

In Lecture 7, we established that numerical stability requires $|G(\xi)| \leq 1$ for all resolvable Fourier phase angles $\xi = k\Delta x \in [-\pi, \pi]$. Under this criterion, the First-Order Upwind (FOU), Lax-Friedrichs (LxF), and Lax-Wendroff (LW) schemes are all stable when the CFL condition $|\nu| = \frac{|a|\Delta t}{\Delta x} \leq 1$ is satisfied.

However, stability alone does not tell the full story of how a scheme behaves in practice. When we simulate waves on a computer, two striking artifacts appear:

- **Numerical Dissipation (Diffusion):** First-order schemes damp out peak amplitudes and severely smear sharp fronts over time.
- **Numerical Dispersion:** Higher-order schemes preserve peak amplitudes better, but generate spurious post-shock or pre-shock wiggles (oscillations) near steep gradients.

In this lecture, we introduce **Modified Equation Analysis**, a powerful mathematical tool that reveals the exact continuous partial differential equation our discrete algebraic algorithm is *actually* solving. We will directly link the even-order and odd-order spatial error derivatives to numerical dissipation and dispersion observed in computer simulations, compare classical schemes, and state **Godunov's Theorem**.

2 Modified Equation Analysis

When deriving a finite difference method via Taylor series, we truncate higher-order terms. **Modified equation analysis** reverses this perspective: we substitute Taylor series expansions of all discrete terms back into the difference scheme, express higher time derivatives in terms of spatial derivatives using the differential equation itself, and determine the effective PDE satisfied by the numerical solution up to higher order.

2.1 Modified Equation for the First-Order Upwind Scheme

Recall the First-Order Upwind scheme for $a > 0$:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + a\frac{u_j^n - u_{j-1}^n}{\Delta x} = 0 \tag{1}$$

Expanding $u_j^{n+1} = u(x_j, t^n + \Delta t)$ and $u_{j-1}^n = u(x_j - \Delta x, t^n)$ in Taylor series:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \frac{\partial u}{\partial t} + \frac{\Delta t}{2}\frac{\partial^2 u}{\partial t^2} + \frac{\Delta t^2}{6}\frac{\partial^3 u}{\partial t^3} + \dots \tag{2}$$

$$\frac{u_j^n - u_{j-1}^n}{\Delta x} = \frac{\partial u}{\partial x} - \frac{\Delta x}{2}\frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^2}{6}\frac{\partial^3 u}{\partial x^3} - \dots \tag{3}$$

Substituting equations (**??**) and (**??**) into (**??**):

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = -\frac{\Delta t}{2}\frac{\partial^2 u}{\partial t^2} + \frac{a\Delta x}{2}\frac{\partial^2 u}{\partial x^2} - \frac{\Delta t^2}{6}\frac{\partial^3 u}{\partial t^3} - \frac{a\Delta x^2}{6}\frac{\partial^3 u}{\partial x^3} + \dots \tag{4}$$

2.1.1 Eliminating Time Derivatives Using the Governing PDE

To express the right-hand side of equation (??) purely in terms of spatial derivatives, we use the governing PDE itself ($\frac{\partial u}{\partial t} = -a\frac{\partial u}{\partial x}$). Differentiating both sides with respect to time:

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial t}\left(\frac{\partial u}{\partial t}\right) = \frac{\partial}{\partial t}\left(-a\frac{\partial u}{\partial x}\right) = -a\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial t}\right) = -a\frac{\partial}{\partial x}\left(-a\frac{\partial u}{\partial x}\right) = a^2\frac{\partial^2 u}{\partial x^2} \tag{5}$$

$$\frac{\partial^3 u}{\partial t^3} = \frac{\partial}{\partial t}\left(a^2\frac{\partial^2 u}{\partial x^2}\right) = a^2\frac{\partial^2}{\partial x^2}\left(\frac{\partial u}{\partial t}\right) = a^2\frac{\partial^2}{\partial x^2}\left(-a\frac{\partial u}{\partial x}\right) = -a^3\frac{\partial^3 u}{\partial x^3} \tag{6}$$

Substituting these into equation (??) and grouping powers:

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = \left(\frac{a\Delta x}{2} - \frac{a^2\Delta t}{2}\right)\frac{\partial^2 u}{\partial x^2} - \left(\frac{a\Delta x^2}{6} - \frac{a^3\Delta t^2}{6}\right)\frac{\partial^3 u}{\partial x^3} + \dots \tag{7}$$

Factoring out the CFL number $\nu = \frac{a\Delta t}{\Delta x}$:

$$\boxed{\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = \underbrace{\frac{a\Delta x}{2}(1-\nu)}_{\text{Numerical Dissipation } \alpha_{\text{num}}}\frac{\partial^2 u}{\partial x^2} - \underbrace{\frac{a\Delta x^2}{6}(1-\nu^2)}_{\text{Numerical Dispersion } \beta_{\text{num}}}\frac{\partial^3 u}{\partial x^3} + \dots} \tag{8}$$

Physical Insights from the Modified Equation

- **Even spatial derivative ($\frac{\partial^2 u}{\partial x^2}$):** Acts like physical viscosity or heat conduction. When $\nu < 1$, the coefficient $\alpha_{\text{num}} = \frac{a\Delta x}{2}(1-\nu) > 0$, causing wave amplitudes to decay and sharp fronts to smear.
- **Odd spatial derivative ($\frac{\partial^3 u}{\partial x^3}$):** Acts like a dispersive wave term (similar to the Korteweg-de Vries equation for water waves). It causes different Fourier frequencies to travel at different speeds.
- **Magic CFL number ($\nu = 1$):** When $\nu = 1$, all error coefficients $\alpha_{\text{num}} = \beta_{\text{num}} = 0$ vanish! The numerical solution shifts exactly along the characteristic: $u_j^{n+1} = u_{j-1}^n$.

2.2 Modified Equation for the Lax-Wendroff Scheme

The Lax-Wendroff scheme was deliberately designed to eliminate the second-order diffusion error ($\mathcal{O}(\Delta x)$). Following the exact same expansion procedure for equation (25) of Lecture 7, we find that the first non-vanishing error term is of third order:

$$\boxed{\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = -\underbrace{\frac{a\Delta x^2}{6}(1-\nu^2)}_{\text{Leading Dispersive Error}}\frac{\partial^3 u}{\partial x^3} - \underbrace{\frac{a^2\Delta t\Delta x^2}{8}(1-\nu^2)}_{\text{Fourth-Order Dissipation}}\frac{\partial^4 u}{\partial x^4} + \dots} \tag{9}$$

Key distinction: Lax-Wendroff has **zero second-order artificial viscosity** ($\alpha_{\text{num}} = 0$). Its leading error is purely **dispersive** ($\sim \frac{\partial^3 u}{\partial x^3}$). This explains why Lax-Wendroff does not smear smooth waves, but produces oscillations near discontinuities!

3 Numerical Simulations: Smooth Waves vs. Discontinuous Fronts

To directly observe how the numerical dissipation ($\frac{\partial^2 u}{\partial x^2}$) and numerical dispersion ($\frac{\partial^3 u}{\partial x^3}$) identified in our modified equations manifest in practice, we simulate both a smooth Gaussian pulse and a discontinuous square wave under periodic boundary conditions using Julia.

From Figure ??, we observe:

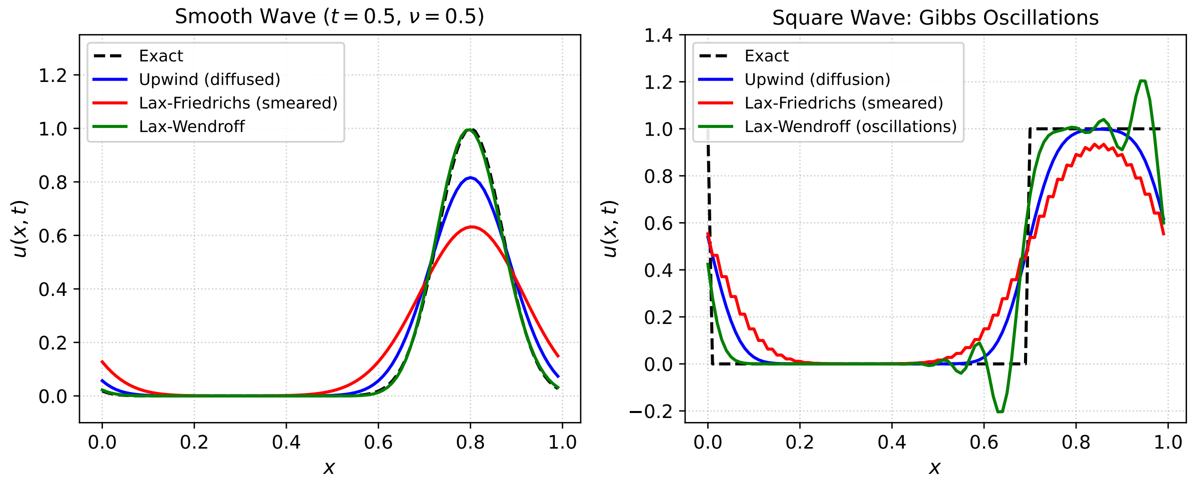

Figure 1: Advection simulation ($a = 1.0, \nu = 0.5$, grid $N = 100$). Left: Smooth Gaussian wave after half transit. Right: Discontinuous square wave showing heavy diffusion (Upwind, Lax-Friedrichs) vs dispersive post-shock oscillations (Lax-Wendroff). Generated with Julia.

1. **Upwind (FOU):** Monotonic (no oscillations), but broadens the pulse significantly due to $\mathcal{O}(\Delta x)$ numerical diffusion.
2. **Lax-Friedrichs:** Massive numerical diffusion ($\sim \frac{\Delta x^2}{2\Delta t}$); the square wave is completely degraded into a blurred hump.
3. **Lax-Wendroff:** High resolution and sharp gradient, but prominent **Gibbs-like oscillations** trail behind the discontinuous jumps. Crucially, the solution undershoots ($u < 0$) and overshoots ($u > 1$), creating non-physical extrema!

4 Godunov's Barrier Theorem

This dilemma between excessive smearing (first-order schemes) and unphysical oscillations (second-order schemes) is not accidental. In 1959, Sergei Godunov proved a landmark impossibility result in numerical analysis:

Godunov's Theorem (1959)

Linear numerical schemes for solving partial differential equations cannot be simultaneously higher than first-order accurate and monotonicity-preserving (oscillation-free).

Implication for CFD: If we restrict ourselves to linear schemes with constant coefficients, we are forced to choose between the heavy blur of first-order schemes or the spurious oscillations of second-order schemes. To overcome this fundamental barrier, modern CFD relies on **non-linear high-resolution methods** (such as TVD schemes, flux limiters, and ENO/WENO methods, which we will study in Module 4).

5 Exercises

Exercises for Lecture 8

E1. Modified Equation for Lax-Friedrichs: Starting from the Lax-Friedrichs difference equation:

$$\frac{u_j^{n+1} - \frac{1}{2}(u_{j+1}^n + u_{j-1}^n)}{\Delta t} + a\frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0$$

use Taylor expansions and substitute time derivatives with spatial derivatives using $u_t = -au_x$ to derive its modified equation up to second order:

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = \frac{\Delta x^2}{2\Delta t}(1-\nu^2)\frac{\partial^2 u}{\partial x^2}$$

Explain why Lax-Friedrichs is far more diffusive than Upwind when $\nu \ll 1$.

E2. Dispersive Nature of the Third Derivative: In the modified equation for Lax-Wendroff:

$$\frac{\partial u}{\partial t} + a\frac{\partial u}{\partial x} = -\frac{a\Delta x^2}{6}(1-\nu^2)\frac{\partial^3 u}{\partial x^3}$$

consider a harmonic wave $u(x,t) = e^{\mathrm{i}(kx-\omega t)}$. Substitute this wave into the modified equation to determine the dispersion relation $\omega(k)$, and show that the phase velocity is $c(k) = \frac{\omega}{k} = a - \frac{a\Delta x^2}{6}(1-\nu^2)k^2$. Deduce that shorter wavelengths (larger k) travel slower than a, directly explaining why the oscillations in Figure **??** trail **behind** the shock!

E3. Modified Equation for Diffusion Equation: For the heat equation $u_t = \alpha u_{xx}$ discretized by FTCS, derive the modified equation up to fourth spatial order and identify the leading numerical error term.